AWS ENVIRONMENTAL HACKS 2026

FLOODSAFE

SENTINEL PRECISION TRIAGE SYSTEM



This presentation outlines the crisis of post-flood water contamination, the operational bottlenecks in disaster response, and how FloodSafe leverages submodular optimization and AWS to save lives.

DEPLOYMENT PHASE

PHASE_01 // CRISIS_MANAGEMENT

STATUS

MISSION CRITICAL READY

TECHNICAL BRIEFING // WATER SECURITY

The Hidden Killer After Floods



VISUAL INTELLIGENCE

The Visual Trap: Clear-looking water masking lethal pathogen concentrations.

> ***"Receding floodwaters create a lethal, invisible trap by flushing pathogens into community water sources."***

VECTOR // INUNDATION OF INFRASTRUCTURE

Receding floodwaters inundate latrines, sewers, and industrial drains, flushing deadly pathogens directly into community wells and taps.

ANALYSIS // THE VISUAL TRAP

Contaminated floodwater often looks completely clear to families, leading to a false sense of security and continued consumption of untreated water.

IMPACT // LETHAL CONSEQUENCES

Outbreaks of cholera, typhoid, and dysentery historically kill more people than the physical force of the floodwaters themselves.

LOGISTICS DISPARITY ANALYSIS // SECTOR 07

500 Wells vs. 15 Test Kits

500

DRINKING WATER POINTS

15

FIELD TEST KITS

> **"Emergency agencies are crippled by a massive disparity between infrastructure scale and testing capacity."**

THE BOTTLENECK

UNICEF, Red Cross, and NDRF crews face hundreds of points per district, while trucks carry a fraction of necessary chemical kits.

CURRENT FIELD FAILURE

Without intelligent decision-support tools, crews are forced to test blindly based on road accessibility or arbitrary ranking.

REDUNDANCY ISSUES

Testing resources cluster redundantly in accessible areas, leaving isolated, high-risk communities without any validation.

MISSION-CRITICAL TRIAGE ARCHITECTURE

From Guesswork to Precision Public-Health Triage

FloodSafe transforms guesswork into precision triage through space radar, terrain physics, and explainable scoring.

PILLAR_01

Space Radar & Terrain Physics

Utilizes Copernicus Sentinel-1 SAR C-band microwave radar to penetrate storm clouds. DEM Topographic Wetness Index (TWI) calculates downhill

PILLAR_02

Explainable Multi-Criteria Scoring

A transparent 5-factor linear formula with neutral 0.5 imputation. Provides strictly non-diagnostic decision support for coordinators, ensuring every triage

PILLAR_03

Submodular Kit Optimization

Mathematically eliminates redundant overlapping tests. The algorithm optimizes logistics to ensure the maximum number of unique human lives

TECHNICAL PERFORMANCE METRICS

Submodular Optimization: +194.1% Greater Population Coverage

IMPACT GAIN

+194.1%

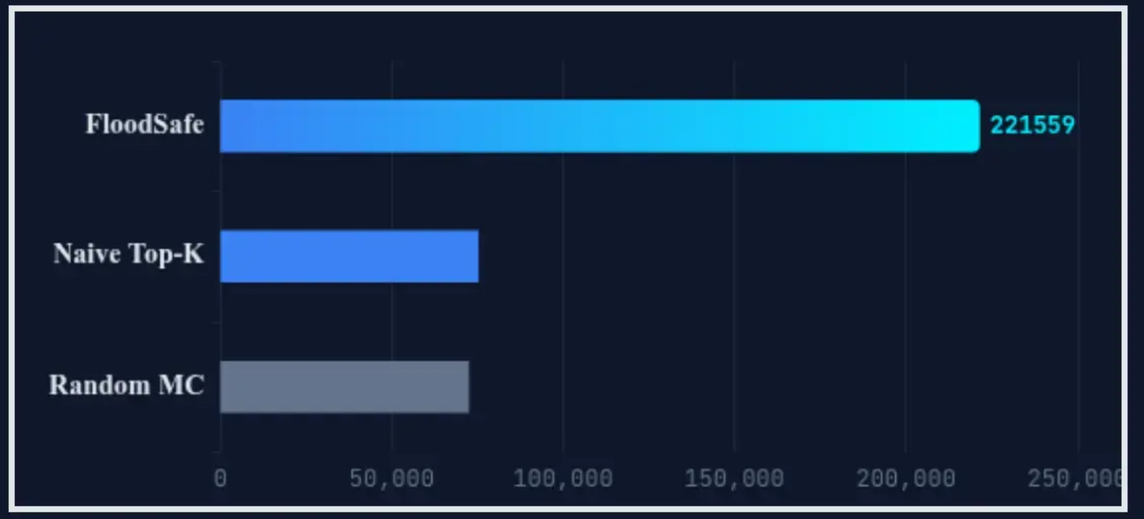
FloodSafe disperses kits across separate at-risk communities, eliminating the redundant overlap inherent in naive clustering.

EVALUATION PARAMETERS

Candidates 500 Units	Budget (K) 15 Kits
Radius (R) 500 Meters	Runtime 24.6ms

Population Coverage Comparison

Total unique people covered across different allocation strategies.



VS. NAIVE TOP-K

75,341 Coverage

+146,218 Gain

VS. RANDOM MONTE CARLO

72,575 Coverage

+148,984 Gain

INFRASTRUCTURE & SECURITY

AWS Serverless Architecture & Cedar Policy Guardrails

— AWS Cloud Stack



STORAGE LAYER

Persistent tables for sources, testing plans, and immutable audit logs.

INFRASTRUCTURE

API Gateway & AWS SAM for robust blueprinting and deployment.

— Security Guardrails

AWS Cedar Engine

Zero-Trust RBAC
Strict default-deny Role-Based Access Control across all services.

MISSION-CRITICAL INVARIANT

"Software is strictly FORBIDDEN from issuing automated public panic advisories without authorized statutory sign-off."

REAL-TIME PHYSICAL GROUND TRUTH

In-Situ Sensing Station & Interactive Wokwi Simulation

> *"In-situ sensing bridges the satellite orbit gap with real-time physical ground truth."*

HARDWARE NODE ARCHITECTURE

- ESP32 DevKit Core + HC-SR04 Ultrasonic Transducer
- SSD1306 OLED Telemetry Display & Piezo Alarm
- Tri-color Status LED visual triage indicators

OPERATIONAL CONTINUITY

Bridges the **6-12 day** satellite revisit gap with **minute-by-minute** continuous monitoring.

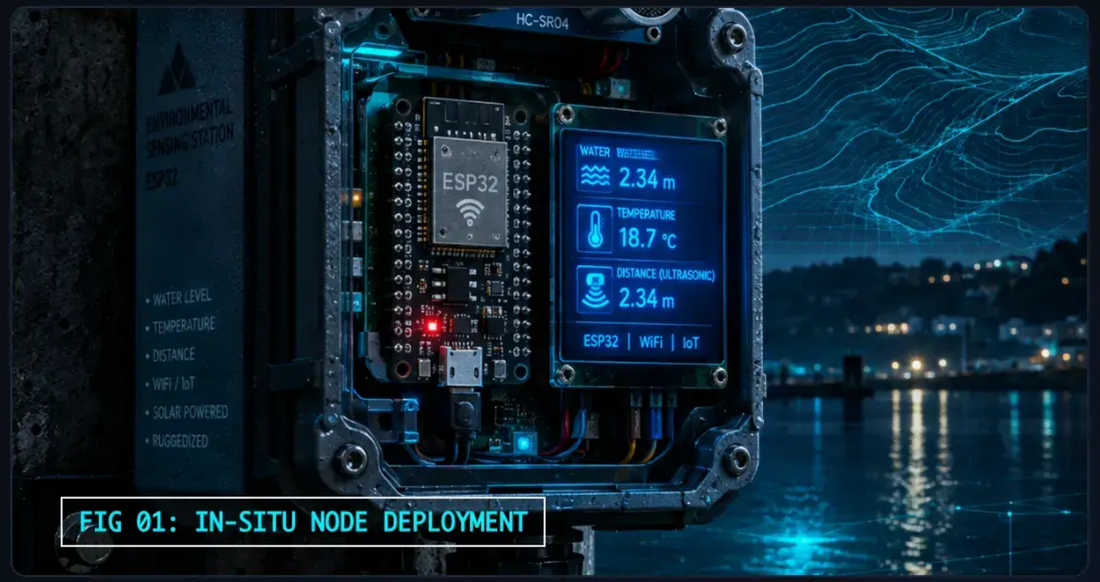


FIG 01: IN-SITU NODE DEPLOYMENT

INTERACTIVE SIMULATION

RESTful JSON STREAM

Live browser-based testing at `wokwi.com/projects/...` featuring an interactive depth slider for real-time logic validation.

SIMULATION ENVIRONMENT
WOKWI V2.0 INTERACTIVE

STATUS
LIVE TELEMETRY

ENGINEERING HEALTH & DEPLOYMENT AUDIT

Production-Ready & Fully Verified

MODULE_01 // IMPACT

LIVE_STATUS

Idea & Human Protection

Tackles the #1 post-flood killer with a scientifically validated intervention model.

MODULE_02 // INFRASTRUCTURE

Cloud Architecture

- Real Cedar RBAC Security Layer
- DynamoDB High-Performance Tables
- SAM-deployable Lambda Functions

MODULE_03 // UX_ENGINE

Usability & GIS Interface

Dark-mode Leaflet GIS canvas featuring a transparent **'WHY?'** mathematical rationale panel.

`Interactive Cedar Studio` `High-Contrast Mapping`

MODULE_04 // INTEGRITY

VERIFIED_OK

Engineering Health

45/45
PYTEST TESTS PASSED

10/10
SPEC INTEGRITY CHECKS

MISSION COMPLETE

THANK YOU

FloodSafe is ready for deployment. Access the live demo and source code below to explore the future of disaster response.

LIVE SYSTEM

Live Demo

flood-safe.onrender.com

SOURCE CODE

GitHub Repository

github.com/SumedhXy/Flood_safe

STATION: COMMAND_CENTRAL ENCRYPTION: AES-256 STATUS: STANDBY

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